DWT-DCT Watermark Design Under AI-Based Image Enhancement

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***Abstract*—Classical DWT-DCT image watermarking has traditionally been evaluated against compression, filtering, and noise-based attacks, where watermark degradation is closely coupled with visible image distortion. We investigate whether AI-based image enhancement introduces a fundamentally different robustness challenge. Across 100 images from the Kodak and TAMPERE17 datasets at four embedding strengths, Real-ESRGAN reduces mean normalized correlation (NC) from 0.999 to 0.795 while maintaining the highest perceptual quality among all watermark-degrading attacks evaluated (SSIM 0.885); ESPCN, FSRCNN, and LapSRN, sharing the identical $4\times$ upscale and Lanczos downscale pipeline, preserve watermark recovery (NC ≥ 0.999), indicating that degradation arises from a specific class of learned reconstruction rather than from resampling or learned super-resolution per se. Coefficient-level analysis across 102,400 DCT blocks reveals markedly different perturbation profiles for Real-ESRGAN and JPEG compression, with strongly opposing coefficient rankings (Spearman $\rho = -0.716$ vs. empirically measured JPEG damage; $\rho = -0.907$ vs. the JPEG quantization table). An exhaustive search over 1,953 coefficient pairs identifies a 37-pair Pareto frontier and establishes that no pair simultaneously optimizes robustness to both attacks. A three-pair head-to-head validation across all 100 images at four embedding strengths confirms strongly attack-specific ordering: the high-frequency pair $(7, 5)/(7, 7)$ achieves the highest Real-ESRGAN NC (0.831 vs. 0.783 for the standard pair), while the analytically balanced pair $(2, 3)/(3, 2)$ achieves the highest JPEG NC (0.998); its Real-ESRGAN comparison is non-final due to an identified and controlled attack-pipeline inconsistency, pending a full rerun. A decision-margin analysis over 307,200 embedded blocks characterizes the mechanism: the perturbation is anti-correlated with the embedded coefficient shift, removing 30–53% of the embedded coefficient differential and moving each block toward its natural coefficient ordering — behavior consistent with an *endogenous restoration* process rather than exogenous noise — and 77–91% of bit flips occur in blocks whose natural ordering opposes the embedded bit. PSNR-oriented super-resolution models show the same perturbation direction at magnitudes an order of magnitude below the enforced decision margin, consistent with why they leave the watermark intact. The ESRGAN/JPEG anti-complementarity pattern is stable across embedding strengths $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$ (Spearman ρ range ≤ 0.010; Jaccard of top-10 damaged positions $= 1.0$). These findings establish that attack-aware coefficient selection is necessary but cannot simultaneously optimize against both AI-enhancement and JPEG attacks within this scheme.**

***Index Terms*—digital watermarking, DWT-DCT, image forensics, robustness evaluation, super-resolution, AI enhancement, Real-ESRGAN, coefficient selection**

I. Introduction

Robustness evaluation for classical image watermarking rests on a quality–attack tradeoff: the standard benchmark attacks — JPEG compression, additive noise, filtering, geometric distortion [1], [2] — all degrade image quality in proportion to the damage they inflict on the watermark. An attacker who wishes to remove a watermark must accept perceptible image damage. This assumption is not incidental; it directly informs how embedding positions and strengths are chosen in deployed systems, and it defines what "robust" has meant in three decades of evaluation practice.

This paper asks a single question: *does consumer AI-based image enhancement violate this assumption for classical DWT-DCT watermarking — and if so, why, and what can be done about it within the classical scheme?*

The question is practically urgent. Combined DWT-DCT watermarking [3] remains a canonical choice in cameras, broadcast equipment, and digital asset management, valued for computational simplicity and independence from machine-learning infrastructure, and it continues to serve as the classical baseline in watermarking research [4]. Meanwhile, super-resolution models such as Real-ESRGAN [5] are embedded in consumer photo tools. A user who "enhances" a watermarked photograph may degrade the embedded signal with no forensic intent and — critically — no visible indication that anything was altered. Whether this constitutes a meaningful threat to deployed classical schemes has not been systematically studied: the WAVES benchmark [4] evaluated DWT-DCT only in an appendix and included no enhancement-class attacks; W-Bench [6] evaluates only neural watermarks.

We answer the question in four stages, which structure the paper. *Observation* (Section V): Real-ESRGAN reduces mean NC to 0.795 while achieving SSIM of 0.885, the highest perceptual quality of any watermark-degrading attack we test — a combination absent from every classical attack profile. ESPCN [7], FSRCNN [8], and LapSRN [9], passed through the identical upscale–downscale pipeline, leave watermarks intact, isolating the cause to the learned reconstruction of the GAN-trained restoration model rather than to learned super-resolution as a class. *Mechanism* (Section VI): a stability analysis over 102,400 DCT blocks shows Real-ESRGAN concentrates damage in the DC coefficient at 5–14$\times$ the magnitude of any other position, whereas JPEG distributes damage nearly uniformly, with opposing coefficient rankings (Spearman $\rho = -0.716$ to -0.907); a decision-margin analysis over 307,200 embedded blocks then shows that Real-ESRGAN's perturbation is anti-correlated with the embedded coefficient shift (r up to -0.80), removing 30–53% of the embedded differential and moving each block toward its natural coefficient ordering — behavior consistent with an *endogenous* restoration process, in contrast to JPEG's exogenous quantization noise. *Design implication and validation* (Section VI): an exhaustive search over 1,953 candidate pairs identifies the ESRGAN/JPEG Pareto frontier and the attack-specific structure of optimal positions; three-pair full validation on 100 images and four embedding strengths confirms large, statistically significant, attack-dependent NC differences. *Stability* (Section VI): the damage-profile anti-complementarity is stable across all tested embedding strengths (ρ range ≤ 0.010; position rank order perfectly preserved).

Our contributions are:

- To the best of our knowledge, the first systematic evaluation of a classical DWT-DCT scheme under learned super-resolution, demonstrating that Real-ESRGAN exhibits a quality–robustness profile unreachable by classical attacks, with three PSNR-oriented models (ESPCN, FSRCNN, LapSRN) as pipeline controls attributing the effect to GAN-trained restoration rather than to learned super-resolution as a class.
- A mechanistic characterization showing Real-ESRGAN perturbs the DCT domain in a fundamentally different pattern than JPEG (DC-concentrated vs. uniform; $\rho \approx -0.9$ opposing rankings), which invalidates the coefficient-selection assumptions inherited from the compression era.
- A block-level decision-margin analysis showing that Real-ESRGAN's perturbation behaves as an *endogenous, signal-dependent* attack: it selectively removes the coefficient differential the embedder inserted against natural-image statistics, concentrating failures (risk ratio 3.4–9.5×) in blocks whose natural coefficient ordering opposes the embedded bit. The model explains per-pair flip rates to within 1 pp, explains why the high-frequency pair survives best (near-zero natural margins), and yields a new position-selection rule: minimize the natural margin $|M_{\mathrm{orig}}|$, not per-position damage.
- An exhaustive 1,953-pair Pareto analysis identifying the structure of the ESRGAN/JPEG robustness tradeoff, with full empirical validation showing strongly attack-dependent NC ordering: no coefficient pair simultaneously optimizes against both AI-enhancement and JPEG attacks.
- An embedding-strength stability result showing the ESRGAN/JPEG damage-profile anti-complementarity is stable across $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$, indicating it is a structural property of the attacks rather than a tuning artifact.

Section II reviews related work. Section III summarizes the DWT-DCT scheme. Section IV describes the experimental setup. Sections V and VI present the observation, mechanism, and validation. Section VII concludes.

II. Related Work

Transform-domain methods dominate invisible watermarking [2], embedding in DCT [10], DWT [11], or combined coefficients; Al-Haj [3] introduced the DWT-DCT scheme evaluated here, with later variants adding SVD [12] or optimized coefficient selection [13]. Across this lineage, robustness evaluation follows the classical protocol of compression, noise, filtering, and geometric attacks [1]; we are not aware of prior work evaluating classical frequency-domain watermarks against AI-based enhancement. The models we test span consumer-deployed GAN-trained blind restoration (Real-ESRGAN [5]) and PSNR-oriented CNN super-resolvers trained against bicubic downsampling (ESPCN [7], FSRCNN [8], LapSRN [9]); unlike regeneration attacks, these refine images in place while restructuring frequency content.

On the neural side, HiDDeN [14] and StegaStamp [15] established learned encoder–decoder watermarking; W-Bench/VINE [6] hardens neural watermarks against generative editing and SuperMark [16] uses diffusion super-resolution as an embedding tool, while Zhao et al. [17] showed diffusion regeneration removes neural watermarks. The WAVES benchmark [4] systematized attacks for neural schemes but evaluated DWT-DCT only in an appendix, without enhancement-class attacks; the present work addresses that gap for classical schemes.

III. DWT-DCT Watermarking

A. Scheme

The scheme is blind: extraction requires only the secret key, no reference image [2]. The host image is converted from RGB to YCbCr (ITU-R BT.601) and only the luminance channel Y is modified. A single-level Haar DWT of Y yields subbands LL/LH/HL/HH; the LL subband is partitioned into non-overlapping 8×8 blocks, each encoding one bit $b \in \{0,1\}$ via 2D DCT by enforcing a relative ordering between the coefficients at positions (r_1, c_1) and (r_2, c_2) with margin δ:

$$b = 1 : C(r_1, c_1) \leftarrow \max\big(C(r_1, c_1),\, C(r_2, c_2) + \delta\big) \quad (1)$$

$$b = 0 : C(r_2, c_2) \leftarrow \max\big(C(r_2, c_2),\, C(r_1, c_1) + \delta\big) \quad (2)$$

Extraction applies the same transforms to the (possibly attacked) image and reads $\hat{b} = 1$ if $C(r_1, c_1) > C(r_2, c_2)$, else $\hat{b} = 0$.

B. Parameters and Metrics

Baseline positions are $(r_1, c_1) = (4, 1)$, $(r_2, c_2) = (1, 4)$ with $\delta = 30.0$; a 1024-bit (32×32) watermark is embedded with fixed seed 42. Four strength multipliers $\alpha \in \{0.05, 0.1, 0.2, 0.3\}$ are evaluated. Recovery is measured by normalized correlation (NC) between the original watermark w and extracted $\hat{w}$, both mapped to $\{-1, +1\}$:

$$\mathrm{NC} = \frac{w \cdot \hat{w}}{\|w\|\,\|\hat{w}\|} \quad (3)$$

NC $= 1.0$ indicates perfect recovery; NC ≈ 0 indicates chance-level extraction.

IV. Experimental Setup

A. Dataset

100 images: 24 from the Kodak suite [18] and 76 from the TAMPERE17 noise-free database [19], all resized to 512×512 RGB.

B. Attack Conditions

Classical attacks. JPEG quality 50 (JPEG-Q50), Gaussian noise ($\sigma = 10$), 3×3 median filter, 5×5 Gaussian blur, and 80% crop with resize to 512×512.

AI enhancement attacks. Real-ESRGAN [5], ESPCN [7], FSRCNN [8], and LapSRN [9]. All upscale watermarked images $4\times$ to 2048×2048 and resize back to 512×512 via Lanczos resampling. Because all attacks share this identical wrapper and differ only in the enhancement network, the three PSNR-oriented models serve as *pipeline controls*: differences in watermark damage are attributable to the learned reconstruction, not to resampling.

C. Coefficient Pair Analysis

DCT stability analysis. We apply DWT and 8×8 block DCT to each watermarked image and its Real-ESRGAN-enhanced counterpart at $\alpha = 0.1$, computing mean absolute coefficient difference per position across 102,400 blocks total; repeated for JPEG-Q50.

Exhaustive pair search. We test all 1,953 valid coefficient pair configurations (positions from the upper-left 7×7 submatrix of the 8×8 DCT block, excluding the DC coefficient and requiring pair asymmetry). Each pair is evaluated for estimated ESRGAN robustness (E-score, based on damage profile) and JPEG robustness (J-score, based on JPEG quantization step). Pareto-optimal pairs are identified across both objectives.

Three-pair full validation. We select three pairs spanning the Pareto frontier for end-to-end head-to-head comparison: the standard baseline pair $(4, 1)/(1, 4)$, the analytically Pareto-optimal balanced pair $(2, 3)/(3, 2)$, and the high-frequency pair $(7, 5)/(7, 7)$ with the strongest predicted ESRGAN robustness. Each is evaluated across all 100 images and four α values under four attacks (none, Real-ESRGAN, JPEG-Q50, Gaussian blur). Pairwise NC differences are assessed with Wilcoxon signed-rank tests (Shapiro-Wilk normality pre-check); p-values are adjusted using the Benjamini-Hochberg false discovery rate (BH-FDR) procedure across all 16 test conditions.

D. Metrics and Implementation

NC and BER are computed on the extracted bit-grid; PSNR and SSIM [20] are computed on the full RGB images (channel-averaged SSIM), per image per attack, averaged across 100 images. Implementation uses PyTorch, PyWavelets, and OpenCV on Apple M2 hardware; Real-ESRGAN via the `py-real-esrgan` package; ESPCN via OpenCV DNN Super-Resolution ($4\times$). Code is available at [REPO].

TABLE I
MEAN WATERMARK SURVIVAL AND IMAGE QUALITY BY ATTACK, AVERAGED ACROSS 100 IMAGES AND FOUR EMBEDDING STRENGTHS (STANDARD PAIR). NC AND BER MEASURE WATERMARK RECOVERY; PSNR AND SSIM MEASURE PERCEPTUAL QUALITY RELATIVE TO THE ORIGINAL IMAGE.

Attack	NC	BER	PSNR	SSIM
Watermarked only	0.9998	0.0001	40.4	0.981
Gaussian noise	0.9986	0.0007	27.9	0.702
JPEG-Q50	0.9956	0.0022	30.2	0.861
Gaussian blur	0.9215	0.0392	28.7	0.843
Median filter	0.8916	0.0542	29.7	0.849
Crop-Resize	−0.0097	0.5049	14.1	0.278
ESPCN $4\times$	0.9993	0.0004	39.4	0.978
FSRCNN $4\times$	0.9996	0.0002	38.9	0.977
LapSRN $4\times$	0.9994	0.0003	39.1	0.978
Real-ESRGAN $4\times$	0.7948	0.1026	29.2	0.885

V. Results: The Quality–Watermark Asymmetry

A. Watermark Survival Under Attack

Table I reports mean NC, BER, PSNR, and SSIM. Classical signal-processing attacks leave watermarks largely intact: Gaussian noise and JPEG-Q50 produce NC above 0.99, consistent with prior DWT-DCT evaluations [3]; blur and median filtering cause moderate degradation (0.922, 0.892); Crop-Resize destroys the watermark (NC ≈ -0.01, chance-level), consistent with the known vulnerability of block-aligned schemes to geometric desynchronization.

The enhancement attacks diverge sharply. All three PSNR-oriented super-resolution models leave watermarks indistinguishable from the unattacked baseline: ESPCN, FSRCNN, and LapSRN achieve NC ≥ 0.999 at every embedding strength (FSRCNN 0.9996, LapSRN 0.9994 pooled over 400 image–strength combinations each). Real-ESRGAN — the only GAN-trained blind-restoration model in the set — reduces mean NC to 0.795 with BER 0.103 — more destructive than any classical attack except Crop-Resize — while simultaneously achieving SSIM of 0.885, the highest perceptual quality of any watermark-degrading attack in our evaluation. Because all four models share the identical resampling wrapper, this contrast indicates the damage arises from Real-ESRGAN's learned reconstruction: AI enhancement is not inherently threatening, and learned super-resolution as a class is not the threat; the threat is specific to restoration-oriented, GAN-trained architectures.

B. The Quality–Watermark Asymmetry

Figure 1 shows the central observation. Every classical attack that reduces NC also reduces SSIM: watermark damage is coupled to visible quality damage. Real-ESRGAN decouples them, occupying a high-quality, low-detection region that no classical attack reaches. Operationally, a content owner monitoring watermark integrity would observe degraded detection with no visible indication that the image had been modified — a qualitatively distinct threat from classical attacks.

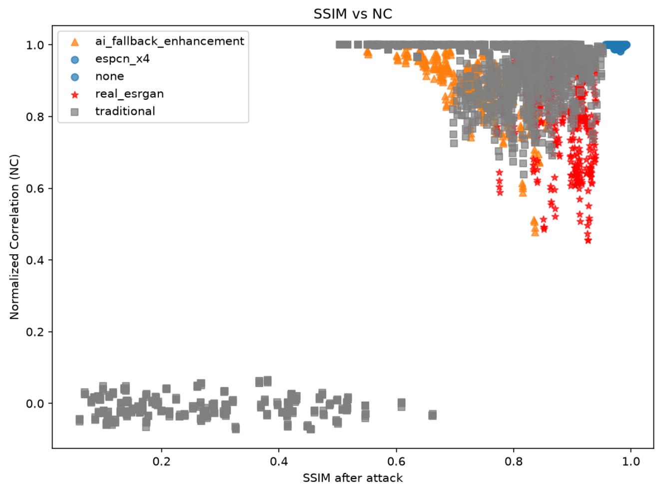

Fig. 1. SSIM vs. NC per attacked image. Real-ESRGAN occupies a high-SSIM, low-NC region unreachable by any classical attack; ESPCN clusters near NC $= 1.0$ at high SSIM.

TABLE II
MEAN NC UNDER REAL-ESRGAN BY EMBEDDING STRENGTH α, STANDARD PAIR $(4,1)/(1,4)$, AVERAGED ACROSS 100 IMAGES.

α	NC (Real-ESRGAN)	SSIM
0.05	0.774	0.886
0.10	0.783	0.885
0.20	0.802	0.885
0.30	0.820	0.884

C. *Embedding Strength Does Not Restore Robustness*

Table II shows NC improves only marginally ($+0.046$) as α increases sixfold from 0.05 to 0.30, while SSIM remains flat. For classical attacks, stronger embedding produces proportionally better survival; Real-ESRGAN does not exhibit this behavior, consistent with the model overwriting DCT coefficient relationships during reconstruction rather than merely attenuating the embedded signal.

VI. MECHANISM AND MITIGATION

A. *DCT Coefficient Damage Profiles*

Figure 2 shows mean absolute perturbation per DCT position. Real-ESRGAN concentrates damage almost entirely in the DC coefficient at $(0,0)$: its mean perturbation is $5.3\times$ that of the most-damaged non-DC position and $14\times$ the median non-DC position, with elevated perturbation extending into the low- and mid-frequency region. This pattern is consistent with the learned reconstruction re-rendering local luminance statistics; it cannot be attributed to resampling, since ESPCN shares that pipeline and leaves extraction nearly perfect. JPEG-Q50 distributes damage across all 64 positions in a narrow range with mild elevation at higher frequencies, consistent with its quantization table design.

The two profiles are strongly anti-complementary: Spearman rank correlation between the per-position ESRGAN and JPEG damage profiles, measured on the α-resolved profiles of the embedding-strength stability analysis below, is $\rho = -0.716$ ($p < 10^{-10}$), and between the ESRGAN damage profile and the JPEG quantization step-size vector is $\rho = -0.907$ ($p < 10^{-25}$). Positions most damaged by ESRGAN tend to be positions least damaged by JPEG, and vice versa. This anti-complementarity means that for any given pair $(r_1, c_1)/(r_2, c_2)$, improving robustness to one attack by choosing positions where it causes less relative damage tends to hurt robustness to the other attack.

The classical embedding positions $(4,1)$ and $(1,4)$ sit in a mid-frequency zone where Real-ESRGAN's perturbation is elevated relative to the most stable high-frequency corner positions. The coefficient-selection assumptions inherited from the compression era do not transfer to the enhancement threat.

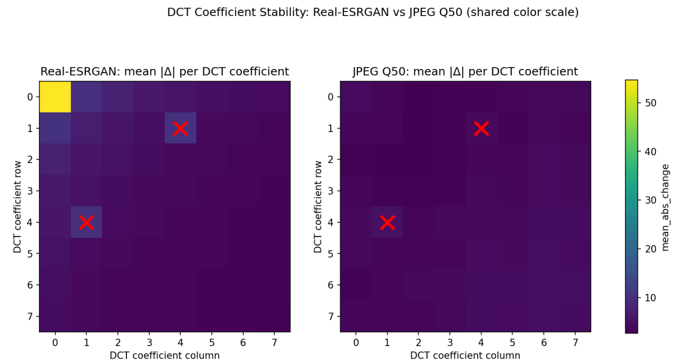

Fig. 2. Mean absolute DCT coefficient perturbation per position for Real-ESRGAN (left) and JPEG-Q50 (right) across 102,400 blocks, shared color scale. Real-ESRGAN concentrates damage in the DC coefficient; JPEG distributes damage nearly uniformly. Baseline embedding positions $(4,1)$ and $(1,4)$ are marked.

B. *Exhaustive Pair Search and Pareto Frontier*

To characterize the structure of the ESRGAN/JPEG tradeoff, we compute robustness scores for all 1,953 valid coefficient pairs derived from the upper-left 7×7 submatrix of the DCT block. For each pair, an E-score summarizes predicted ESRGAN stability (based on the damage profile) and a J-score summarizes predicted JPEG robustness (based on the JPEG quantization table). The Pareto frontier — pairs no other configuration dominates on both objectives simultaneously — contains 37 pairs out of 1,953. The standard baseline pair $(4,1)/(1,4)$ lies off the Pareto frontier (E-score $= 0.028$, J-score $= 0.877$), confirming it is not optimal under either attack in isolation or their combination.

The anti-complementarity of the damage profiles is directly reflected in the Pareto plot (Figure 3): the ESRGAN NC and JPEG NC of individual pairs are negatively correlated (Spearman $\rho = -0.488$, $p < 0.001$) across the 43 screened pairs. No pair improves against both attacks simultaneously relative to the baseline.

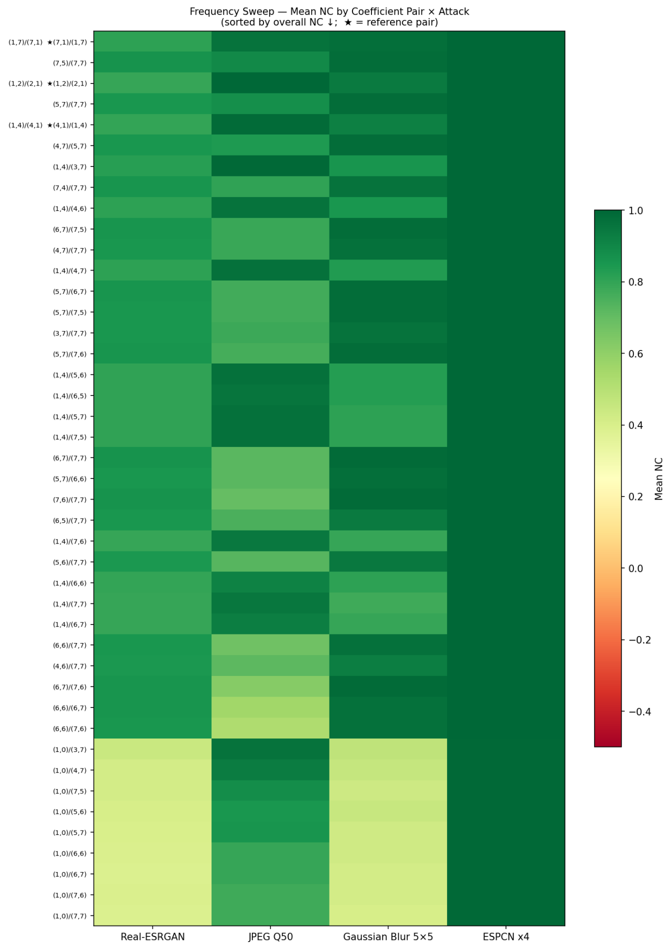

Fig. 3. Real-ESRGAN NC vs. JPEG-Q50 NC per coefficient pair. The Pareto frontier (upper-right boundary) shows no pair simultaneously maximizes both. Pairs anchored near the high-frequency corner $(7, 7)$ achieve highest ESRGAN NC; pairs in the mid-frequency band achieve highest JPEG NC.

C. Three-Pair Full Validation

We select three pairs spanning the Pareto frontier for end-to-end validation: the standard pair $(4, 1)/(1, 4)$, the analytically Pareto-optimal pair $(2, 3)/(3, 2)$ (highest combined E+J score among the 37 Pareto-optimal pairs), and the high-frequency pair $(7, 5)/(7, 7)$ (highest predicted ESRGAN robustness). Each pair is evaluated across all 100 images at all four α values under four attack conditions. Table III shows results at $\alpha = 0.10$.

Real-ESRGAN. The HF pair achieves NC of 0.831, 0.048 above standard ($p < 10^{-13}$, $d = 0.89$), a difference stable across all four embedding strengths. The balanced pair's tabulated value (0.683) must be read with care: its Real-ESRGAN images were generated by a patch-based invocation of the attack, whereas the standard and HF images used the whole-image invocation, and a control experiment shows the patch-based pipeline alone is substantially harsher (running the *standard* pair through it costs -0.102 NC on the same watermarked images; $n = 20$, Wilcoxon $p = 10^{-4}$) — almost exactly the apparent balanced–standard gap. This pipeline inconsistency in the original validation run was identified by the decision-margin control experiment of Section VI-D. Under a like-for-like pipeline the balanced pair does *not* underperform the standard pair; it is marginally better ($+0.020$, $n = 20$, $p = 0.038$), with HF best under both pipelines. We therefore do not present the balanced-pair Real-ESRGAN comparison as a final result: a like-for-like rerun of the balanced arm on all 100 images is required before camera-ready, and the tabulated 0.683 should be read only as an upper bound on the harsher pipeline's effect. What can be concluded already is that the analytically predicted *large* advantage of the balanced pair fails to materialize under either pipeline; the next subsection diagnoses why.

TABLE III

MEAN NC $\pm$ STANDARD DEVIATION AT $\alpha = 0.10$ ACROSS 100 IMAGES FOR THREE COEFFICIENT PAIRS UNDER FOUR ATTACK CONDITIONS. ALL PAIRWISE NC DIFFERENCES ACROSS ALL ATTACK/PAIR COMBINATIONS ARE STATISTICALLY SIGNIFICANT AFTER BH–FDR CORRECTION EXCEPT NO-ATTACK COMPARISONS ($p < 10^{-12}$ OR N.S. AS INDICATED). †THE BALANCED PAIR'S REAL-ESRGAN IMAGES WERE PRODUCED BY A PATCH-BASED INVOCATION OF THE ATTACK THAT IS MEASURABLY HARSHER THAN THE WHOLE-IMAGE INVOCATION USED FOR THE OTHER TWO PAIRS; SEE THE LIKE-FOR-LIKE CONTROL IN THE TEXT BEFORE COMPARING THIS VALUE ACROSS COLUMNS.

Attack	Standard $(4, 1)/(1, 4)$	Balanced $(2, 3)/(3, 2)$	HF $(7, 5)/(7, 7)$
No attack	0.9997 ± 0.000	0.9997 ± 0.000	1.000 ± 0.000
Real-ESRGAN	0.783 ± 0.124	$0.683 \pm 0.139^{\dagger}$	0.831 ± 0.109
JPEG-Q50	0.995 ± 0.004	0.998 ± 0.001	0.902 ± 0.018
Gaussian blur	0.910 ± 0.013	0.933 ± 0.011	0.977 ± 0.005
PSNR (dB)	40.6	40.9	43.9

JPEG-Q50. The ordering reverses: Balanced (0.998) $>$ Standard (0.995) $\gg$ HF (0.902). The HF pair suffers a large JPEG regression ($\Delta\mathrm{NC} = -0.094$ vs. standard, $p < 10^{-17}$).

Gaussian blur. Ordering follows ESRGAN: HF (0.977) $>$ Balanced (0.933) $>$ Standard (0.910); all differences are large and highly significant.

Imperceptibility. The balanced pair achieves higher PSNR ($+0.30$ dB at $\alpha = 0.10$), and the HF pair achieves substantially higher PSNR ($+3.3$ dB), eliminating below-threshold imperceptibility failures present in the standard pair.

Key finding. No pair simultaneously achieves the highest NC across all attacks. The analytical Pareto ranking is not preserved in magnitude in end-to-end evaluation: the pair predicted to be the best combined choice (balanced, $(2, 3)/(3, 2)$) delivers only a $+0.02$ like-for-like improvement over the standard pair under Real-ESRGAN ($n = 20$ control; full rerun pending), far short of the analytically predicted gap, while the HF pair — selected by a simpler per-position damage heuristic — outperforms both. Embedding position selection offers attack-aware partial mitigation, but the ESRGAN/JPEG tradeoff cannot be resolved within this scheme.

D. Why the Analytical Ranking Fails: Enhancement as Endogenous Restoration

To diagnose the failure of the damage-based analytical ranking, we measure the detector's decision statistic directly: survival is governed by the signed margin $M = C_1 - C_2$, measured per block before and after Real-ESRGAN for all three pairs at $\alpha = 0.10$ (307,200 blocks), together with each block's *natural* margin M_{orig} in the unwatermarked original. NC loss is entirely margin sign flips (flip and bit-error rates agree to within 0.1%); the embedding's margin floor clamps $|M|$ near 30 for all pairs, and pre-attack margin magnitude predicts nothing (Spearman $\rho \approx -0.07$, n.s.). The best image-level predictor of NC is the standard deviation of the *differential* perturbation $\sigma(\Delta C_1 - \Delta C_2)$ ($\rho = -0.887$), outperforming the absolute per-coefficient damage used by the analytical E-score ($\rho = -0.825$); common-mode perturbation, to which the comparison detector is blind by design, is irrelevant ($\rho = -0.22$).

The perturbation is *signal-dependent*: it is anti-correlated with the embedding-induced margin shift ($r = -0.73/-0.60/-0.80$ for standard/balanced/HF), removes 32%/30%/53% of what the embedder inserted, and adding M_{orig} to a before-only regression of the post-attack margin raises R^2 from 0.71 to 0.83 (standard pair). This behavior is inconsistent with fixed, position-dependent noise and is consistent with learned image restoration reducing the artificial coefficient relationships introduced by comparison-based embedding — an *endogenous* restoration process; we characterize the model's input–output behavior only and make no claim about its internal computations.

Consequently, the watermark dies where it fought nature: blocks whose natural ordering opposes the embedded bit flip at 3.4–9.5× the rate of agreeing blocks, and 77–91% of all flips occur in that ≈50% of blocks. This suggests why the HF pair survives best despite the largest restoration fraction (53%): its natural margins are tiny (mean $|M_{\text{orig}}| = 10.1$ vs. ≈ 31 for the other pairs), consistent with a restoration process having little to restore against the embedded ordering. The governing quantity for position selection is thus the *differential perturbation of the embedded coefficient pair*, not per-position exogenous damage. Full derivations and additional analyses appear in the extended version.

E. Generalization Across Super-Resolution Architectures

Is this pressure unique to Real-ESRGAN? Repeating the margin analysis at $\alpha = 0.10$ (standard pair) for the three PSNR-oriented models under the identical attack wrapper (Table IV) shows differential perturbation an order of magnitude below the ≈30-unit enforced margin ($\sigma = 1.8$–3.7 vs. 24.2 for Real-ESRGAN), so essentially no block crosses the decision boundary. LapSRN is the instructive intermediate case: its perturbation *direction* carries nearly the same restoration signature as Real-ESRGAN ($r_{\text{undo}} = -0.69$ vs. -0.73) at 7× weaker strength ($k = 4.7\%$ vs. 32.3%), and the few flips these models do cause occur exclusively in blocks whose natural ordering opposes the embedded bit — the same signature, in miniature.

TABLE IV
MARGIN-ANALYSIS MECHANISM QUANTITIES PER SR MODEL (STANDARD PAIR, $\alpha = 0.10$, 102,400 BLOCKS PER MODEL). k = FRACTION OF THE EMBEDDED COEFFICIENT DIFFERENTIAL REMOVED; r_{undo} = CORRELATION BETWEEN DIFFERENTIAL PERTURBATION AND EMBEDDED SHIFT.

Model	Flip rate	$\sigma(\Delta C_1 - \Delta C_2)$	k	r_{undo}
ESPCN 4×	0.02%	1.9	0.5%	−0.14
FSRCNN 4×	0.01%	1.8	0.2%	−0.05
LapSRN 4×	0.02%	3.7	4.7%	−0.69
Real-ESRGAN 4×	10.9%	24.2	32.3%	−0.73

These measurements suggest that a perturbation component directed against the embedded coefficient relationship is present across the learned SR architectures we test, but only the GAN-trained restoration model is strong enough to cross the embedded decision boundary. On current evidence, Real-ESRGAN produces substantial degradation while ESPCN, FSRCNN, and LapSRN preserve the watermark under the same evaluation pipeline: the practical threat is architecture-dependent, not a property of "AI enhancement" as a category.

F. Stability of the Anti-Complementarity Across Embedding Strengths

A key question is whether the ESRGAN/JPEG anti-complementarity depends on the embedding strength α — for example, whether stronger embedding can shift which positions are most vulnerable. We measure the per-position ESRGAN and JPEG DCT damage profiles at all four α values and compute cross-attack Spearman correlations and cross-α rank stability for the standard pair.

The ESRGAN/JPEG profile anti-complementarity is essentially constant across α: ρ ranges from -0.716 at $\alpha = 0.05$ to -0.706 at $\alpha = 0.30$, a total range of 0.010 over a 6× span of embedding strength. The rank ordering of all 64 DCT positions by damage magnitude is also stable: Jaccard similarity of the top-10 most-damaged positions is 1.0 for all cross-α comparisons, for both ESRGAN and JPEG independently. This indicates the anti-complementarity is a structural property of the two attack types, not of the embedding strength. Similarly, Real-ESRGAN damage isotropy (Spearman r between damage at symmetric DCT positions) is stable at $r \approx 0.989$ across all tested α values. The margin analysis of Section VI-D gives this anti-complementarity a mechanistic reading: JPEG is an exogenous quantizer whose damage follows its fixed quantization table, whereas Real-ESRGAN's damage follows the embedded signal itself — consistent with the two attacks being not merely spectrally complementary but mechanistically different kinds of noise, which would explain why no static position choice escapes the trade-off.

Separately, symmetric coefficient pairs achieve mean Real-ESRGAN NC of 0.821 versus 0.689 for asymmetric pairs (43-pair screening set, $\alpha = 0.1$, 20-image subset), disconfirming coordinate symmetry as the cause of Real-ESRGAN vulnerability.

G. Subband Selection Does Not Help

As a supplementary comparison, we evaluated a DWT-SVD scheme embedding in the HH subband at matched imperceptibility (47.7 vs. 48.0 dB PSNR, $\alpha = 0.1$). It achieves NC of only 0.18 under Gaussian blur and JPEG-Q50 and 0.29 under Real-ESRGAN — near-chance extraction under attacks the DWT-DCT-LL scheme survives at 0.93, 0.98, and 0.74 respectively. Moving embedding energy to high-frequency subbands is not a viable escape from the enhancement vulnerability.

H. Limitations

Our evaluation covers four learned super-resolution architectures (ESPCN, FSRCNN, LapSRN, Real-ESRGAN); within this set the destructive behavior is confined to the GAN-trained restoration model, but diffusion-based enhancers and other GAN restorers remain untested, and the margin analysis is reported at a single embedding strength ($\alpha = 0.10$), and the balanced-pair like-for-like control (Section VI-C) remains $n = 20$ pending the camera-ready rerun. All headline results are for one classical scheme (DWT-DCT with ordering-based bit encoding); generalization to other classical schemes remains to be established. The Pareto analysis uses proxy scores (damage-profile-based E-score and J-score) rather than end-to-end NC; the margin analysis of Section VI-D suggests *why* such proxies mispredict for generative attacks (they model a signal-dependent attack as exogenous noise), and the candidate selection rule — minimize the natural margin $|M_{\text{orig}}|$ — has not yet been validated prospectively on new pairs.

VII. Conclusion

We investigated whether AI-based image enhancement violates the quality–attack tradeoff underlying classical watermark robustness evaluation, and found that it can: Real-ESRGAN degrades DWT-DCT watermark recovery to NC $= 0.795$ while producing the highest perceptual quality (SSIM 0.885) of any watermark-degrading attack tested, while ESPCN, FSRCNN, and LapSRN — sharing the identical resampling pipeline — leave watermarks intact. The proximate mechanism is a DC-concentrated perturbation profile fundamentally unlike JPEG's uniform profile, with strongly anti-complementary position rankings ($\rho \approx -0.7$ to -0.9). A 307,200-block decision-margin analysis shows that what governs survival is the *differential perturbation of the embedded coefficient pair*, not absolute coefficient stability: the perturbation removes 30–53% of the embedded coefficient differential and moves each block toward its natural coefficient ordering — consistent with learned restoration reducing the artificial coefficient relationships introduced by comparison-based embedding — so that failures concentrate (risk ratio 3.4–9.5$\times$) where the natural ordering opposes the embedded bit. PSNR-oriented super-resolution models show the same perturbation direction at an order of magnitude lower strength, leaving watermarks intact: on current evidence the threat is architecture-dependent, not a property of learned enhancement as a category. An exhaustive search over 1,953 coefficient pairs confirms that the JPEG/ESRGAN anti-complementarity is reflected in end-to-end NC: no pair simultaneously maximizes Real-ESRGAN and JPEG robustness, and the anti-complementarity is stable across embedding strengths $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$, consistent with its origin in the exogenous-vs-endogenous character of the two attacks rather than in tuning.

Content owners relying on DWT-DCT watermarking should be aware that consumer enhancement tools built on GAN-trained restoration can silently degrade watermark integrity, while common PSNR-oriented super-resolvers do not. Attack-aware coefficient selection offers partial mitigation: the high-frequency pair $(7, 5)/(7, 7)$ improves Real-ESRGAN NC from 0.795 to 0.831 and eliminates imperceptibility failures at the cost of JPEG NC (0.996 to 0.902), and the margin analysis suggests an explanation for its advantage — near-zero natural margins leave a restoration-type attack little to restore — yielding a candidate selection rule (minimize natural $|M_{\text{orig}}|$) in place of per-position damage heuristics. Future work should validate this rule prospectively, extend the evaluation to diffusion-based enhancers and additional GAN restorers, and investigate content-adaptive strategies that jointly optimize coefficient selection and embedding against both exogenous and endogenous attack types.

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